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On estimation of the noise variance in high dimensional probabilistic principal component analysis

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Summary. We develop new statistical theory for probabilistic principal component analysis models in high dimensions. The focus is the estimation of the noise variance, which is an important and unresolved issue when the number of variables is large in comparison with the sample size. We first unveil the reasons for an observed downward bias of the maximum likelihood estimator of the noise variance when the data dimension is high. We then propose a bias-corrected estimator by using random-matrix theory and establish its asymptotic normality. The superiority of the new and bias-corrected estimator over existing alternatives is checked by Monte Carlo experiments with various combinations of (p, n) (the dimension and sample size). Next, we construct a new criterion based on the bias-corrected estimator to determine the number of the principal components, and a consistent estimator is obtained. Its good performance is confirmed by a simulation study and real data analysis. The bias-corrected estimator is also used to derive new asymptotics for the related goodness-of-fit statistic under the high dimensional scheme.

Keywords: Goodness of fit; High dimensional data; Noise variance estimator; Number of principal components; Probabilistic principal component analysis; Random-matrix theory

1. Introduction

Principal component (PC) analysis is a very popular technique in multivariate analysis for dimensionality reduction and feature extraction. Because of dramatic development in data collection technology, high dimensional data are nowadays common in many fields. Natural high dimensional data, such as images, signal processing, documents and biological data, often reside in a low dimensional subspace or low dimensional manifold (Ding *et al.*, 2011). In financial econometrics, it is commonly believed that the variations in a large number of economic variables can be modelled by a small number of reference variables (Forni *et al.*, 2000; Bai and Ng, 2002; Bai, 2003). Consequently, PC analysis is a recommended tool for analysis of such high dimensional data.

There is an underlying probabilistic model behind PC analysis, called probabilistic principal component analysis (PPCA), which is defined as follows. The observation vectors $\{\mathbf{x}_i\}_{1\leqslant i\leqslant n}$ are p dimensional and satisfy the equation

$$\mathbf{x}_i = \boldsymbol{\Lambda}\mathbf{f}_i + \mathbf{e}_i + \boldsymbol{\mu}, \qquad i = 1, \ldots, n. \tag{1}$$

Here, $\mathbf{f}_i$ are m-dimensional *PCs* with $m \ll p$, $\boldsymbol{\Lambda}$ is a $p\times m$ matrix of *loadings* and $\boldsymbol{\mu}$ represents the general mean and $\{\mathbf{e}_i\}_{1\leqslant i\leqslant n}$ are a sequence of independent errors with covariance matrix

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$\mathbf{\Psi}=\sigma^2\mathbf{I}_p$. The parameter σ^2 is the noise variance that we are interested in. None of the quantities on the right-hand side of equation (1) is known or observed (except their sum $\mathbf{x}_i$).

To ensure the identification of the model, constraints must be introduced on the parameters. There are several possibilities for the choice of such constraints; see Table 1 in Bai and Li (2012). A traditional choice is (Anderson (2003), chapter 14)

(a) $\mathbb{E}(\mathbf{f}_i)=\mathbf{0}$ and $\mathbb{E}(\mathbf{f}_i\mathbf{f}_i')=\mathbf{I}$, and
(b) the matrix $\mathbf{\Gamma}:=\mathbf{\Lambda}'\mathbf{\Lambda}$ is diagonal with distinct diagonal elements.

Therefore, the population covariance matrix of $\{\mathbf{x}_i\}_{1\leqslant i\leqslant n}$ is $\mathbf{\Sigma}=\mathbf{\Lambda}\mathbf{\Lambda}'+\sigma^2\mathbf{I}$. Finding a reliable estimator of the noise variance σ^2 is a non-trivial issue for high dimensional data which we now pursue.

The PPCA model (1) can be viewed as a special instance of the approximate factor model (Chamberlain and Rothschild, 1983) where the noise covariance $\mathbf{\Psi}$ can be a general diagonal matrix (the model is also called a strict factor model in statistical literature; see Anderson (2003)). For related recent references on inference of large approximate (or dynamic) factor models, we refer to Bai (2003), Forni *et al.* (2000) and Doz *et al.* (2012).

Let $\bar{\mathbf{x}}$ be the sample mean and define the sample covariance matrix

$$\mathbf{S}_n=\frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})'. \tag{2}$$

Let $\lambda_{n,1}\geqslant\lambda_{n,2}\geqslant\ldots\geqslant\lambda_{n,p}$ be the eigenvalues of $\mathbf{S}_n$. Under the normality assumption on both $\{\mathbf{f}_i\}$ and $\{\mathbf{e}_i\}$, the maximum likelihood estimator of σ^2 is

$$\hat{\sigma}^2=\frac{1}{p-m}\sum_{i=m+1}^{p}\lambda_{n,i}. \tag{3}$$

In the classic setting where the dimension p is relatively small compared with the sample size n (the low dimensional setting), the consistency of $\hat{\sigma}^2$ was established in Anderson and Rubin (1956). Moreover, it is asymptotically normal with the standard root n convergence rate: as $n\rightarrow\infty$,

$$\surd n(\hat{\sigma}^2-\sigma^2)\xrightarrow{\mathcal{D}}\mathcal{N}(0,s^2), \qquad s^2=\frac{2\sigma^4}{p-m}. \tag{4}$$

Actually, Anderson and Amemiya (1988) provided a general central limit theorem (CLT) in an approximate factor model that encompasses the present PPCA model. For the reader's convenience, we provide in the on-line supplement a detailed derivation of result (4) from this general CLT.

The situation is, however, radically different when p is large compared with the sample size n. Recent advances in high dimensional statistics indicate that, in such high dimensional situations, the above asymptotic result is no longer valid and, indeed, it has been reported in the literature that $\hat{\sigma}^2$ seriously underestimates the true noise variance σ^2; see Kritchman and Nadler (2008). However, the exact form of this bias has not been determined (to our best knowledge). Basically, when p becomes large, the sample principal eigenvalues and PCs are no longer consistent estimates of their population counterparts (Baik and Silverstein, 2006; Johnstone and Lu, 2009; Kritchman and Nadler, 2008). Many estimation methods developed in low dimensional settings have been shown to perform poorly even for moderately large p and n (Cragg and Donald, 1997).

As all meaningful inference procedures in the model will unavoidably use some estimator of the noise variance σ^2, such a severe bias needs to be corrected for high dimensional data. Several estimators have been proposed to deal with the high dimensional situation. Kritchman and Nadler (2008) proposed an estimator by solving a system of implicit equations; Ulfarsson and Solo (2008) introduced an estimator using the median of the sample eigenvalues $\{\lambda_{n,i}\}$; Johnstone and Lu (2009) used the median of the sample variances. However, these estimators have been assessed by Monte Carlo experiments only and their theoretical properties have not been investigated.

The main aim of this paper is to provide a new estimator of the noise variance for which rigorous asymptotic theory can be established in the high dimensional setting. First, by using recent advances in random-matrix theory, we found a CLT for the maximum likelihood estimator (MLE) $\hat{\sigma}^2$ in the high dimensional setting. Next, using this identification, we propose a new estimator $\hat{\sigma}^2_*$ for the noise variance by correcting this bias. The asymptotic normality of the new estimator is thus established with explicit asymptotic mean and variance.

Although the asymptotic Gaussian distribution of the new estimator $\hat{\sigma}^2_*$ is established under the high dimensional setting $p \to \infty$, $n \to \infty$ and $p/n \to c > 0$, if we set $c = 0$, i.e. the dimension p is infinitely smaller than n, this Gaussian limit coincides with the classical low dimensional limit given in expression (4). In this sense, the new asymptotic theory extends in a continuous manner the classical low dimensional result to the high dimensional situation. Finite sample properties of the new estimator $\hat{\sigma}^2_*$ have been checked via Monte Carlo experiments in comparison with the above-mentioned four existing estimators. In terms of mean-squared errors and in all the scenarios tested, $\hat{\sigma}^2_*$ outperforms very significantly three of them and is slightly preferable to the other one; see Table 2 in Section 2.1.

To demonstrate further potential benefits of the new estimator $\hat{\sigma}^2_*$, we consider an important inference problem in PPCA, namely the determination of the number of PCs. Bai and Ng (2002) developed six criteria with penalty on both p and n to identify the number of factors in the approximate factor model. The approximate factor model allows the components of the errors $\{\mathbf{e}_i\}$ to be correlated. PPCA can be considered as a simplified instance of this model and, indeed, Bai and Ng (2002) also applied their criteria to PPCA. It is worth noting that the determination of the number m of PCs and the estimation of the noise variance σ^2 are interrelated and Bai and Ng's criteria provide a consistent and joint inference on (m, σ^2) in the high dimensional context. However, this consistency is obtained under the assumption that the variances (or the strengths) of the PCs grow to ∞ with the dimension (see assumption B of Bai and Ng (2002), whereas in our context these variances could be bounded. Therefore, we propose a modified estimator of both (m, σ^2) by implementing our new estimator $\hat{\sigma}^2_*$ in the criteria of Bai and Ng (2002). Furthermore, to deal with possibly bounded variances of PCs, a new penalty function is found based on our new estimator $\hat{\sigma}^2_*$. The resulting procedure provides a consistent joint estimator of (m, σ^2). Moreover, as predicted by our theory, this new procedure has a better performance than Bai and Ng's original procedures in our context with possibly bounded PC variances.

As a final application of the new estimator $\hat{\sigma}^2_*$, we consider the goodness-of-fit test for the PPCA model. The likelihood ratio test (LRT) statistics as well as their classical (low dimensional) χ^2 asymptotic theory have been well known since the work of Amemiya and Anderson (1990). These results are again challenged by high dimensional data and the classical χ^2-limit is no longer valid. We propose a correction to this goodness-of-fit test statistic involving our new estimator $\hat{\sigma}^2_*$ to cope with the high dimensional effects and to establish its asymptotic normality.

The remaining sections are organized as follows. In Section 2, we present the main results of the paper. The new estimator $\hat{\sigma}^2_*$ of the noise variance is proposed first; then a new joint estimator of the number of PCs and the noise variance is constructed. In Section 3, we develop the corrected likelihood ratio test (CLRT) for the goodness of fit of a PPCA model in the high dimensional framework by using the new estimator $\hat{\sigma}^2_*$. Section 4 concludes. The most important technical proofs are gathered in Appendix A whereas the remaining proofs are relegated to the on-line supplementary report. This report contains also many numerical results and additional applications. Lastly, all the code to enable reproduction of the results in Tables 1–7 of the paper and some of the data sets that are used in the paper are available from `http://web.hku.hk/~jeffyao/papersInfo.html`.

2. Main results

The PPCA model (1) is a spiked population model (Johnstone, 2001) since the eigenvalues of the population covariance matrix $\mathbf{\Sigma}$ are

$$\begin{aligned}\operatorname{spec}(\mathbf{\Sigma}) &= (\alpha_1,\ldots,\alpha_m,\underbrace{0,\ldots,0}_{p-m}) + \sigma^2(\underbrace{1,\ldots,1}_{p}) \\ &= \sigma^2(\alpha_1^*,\ldots,\alpha_m^*,\underbrace{1,\ldots,1}_{p-m}),\end{aligned} \tag{5}$$

where $\{\alpha_i\}$ are m non-null eigenvalues of $\mathbf{\Lambda\Lambda}'$ and the notation $\alpha_i^* = \alpha_i/\sigma^2 + 1$ is used. To develop meaningful asymptotic theory in the high dimensional context, we assume that p and n are related so that, when $n\to\infty$, $c_n = p/(n-1)\to c>0$, i.e. p can be large compared with the sample size n and, for the asymptotic theory, p and n tend to ∞ proportionally. Let

$$\phi(\alpha) = \alpha + \frac{c\alpha}{\alpha-1}, \qquad \alpha\neq 1.$$

Following Baik and Silverstein (2006), assume that $\alpha_1^*\geqslant\ldots\geqslant\alpha_m^* > 1+\surd c$, i.e. all the eigenvalues α_i are greater than $\sigma^2\surd c$. It is then known that, for the spiked sample eigenvalues $\{\lambda_{n,i}\}_{1\leqslant i\leqslant m}$ of $\mathbf{S}_n$, almost surely,

$$\lambda_{n,i}\to\sigma^2\phi(\alpha_i^*) = \psi(\sigma^2,c,\alpha_i) = \alpha_i+\sigma^2+\sigma^2 c\left(1+\frac{\sigma^2}{\alpha_i}\right). \tag{6}$$

Moreover, the remaining sample eigenvalues $\{\lambda_{n,i}\}_{m<i\leqslant p}$, called *noise eigenvalues*, will converge to a continuous distribution with support interval $[a(c),b(c)]$ where $a(c)=\sigma^2(1-\surd c)^2$ and $b(c)=\sigma^2(1+\surd c)^2$. In particular, for all $1\leqslant j\leqslant L$ with a prefixed range L and almost surely, $\lambda_{n,m+j}\to b(c)$. It is worth noting that, if $c\to 0$, we recover the low dimensional limits $\lambda_{n,i}\to\alpha_i+\sigma^2$ (population spike eigenvalues) and $\lambda_{n,i}\to\sigma^2$ (population noise eigenvalues) discussed earlier. In addition, the CLT for the spiked eigenvalues was established in Bai and Yao (2008): $\surd n\{\lambda_{n,i}-\sigma^2\phi(\alpha_i^*)\}$ is asymptotically Gaussian.

As explained in Section 1, when the dimension p is large compared with the sample size n, the MLE $\hat{\sigma}^2$ has a negative bias. To identify this bias, we first establish a CLT for $\hat{\sigma}^2$ under the high dimensional scheme.

Theorem 1. Consider the PPCA model (1) with population covariance matrix $\mathbf{\Sigma}=\mathbf{\Lambda\Lambda}'+\sigma^2\mathbf{I}_p$ where both the PCs and the noise are Gaussian. Assume that $p\to\infty$, $n\to\infty$ and $c_n = p/(n-1)\to c>0$, and the non-null eigenvalues of $\mathbf{\Lambda\Lambda}'$ $\{\alpha_i\}$ satisfy $\alpha_i\geqslant\sigma^2\surd c$ $(1\leqslant i\leqslant m)$. Then, we have

$$\frac{p-m}{\sigma^2\surd(2c)}(\hat{\sigma}^2-\sigma^2)+b(\sigma^2)\xrightarrow{\mathcal{D}}\mathcal{N}(0,1),$$

where $b(\sigma^2)=\surd(c/2)\{m+\sigma^2\,\Sigma_{i=1}^m(1/\alpha_i)\}$.

The proof is given in Appendix A. Therefore, for high dimensional data, the MLE $\hat{\sigma}^2$ has an asymptotic bias $-b(\sigma^2)$ (after normalization). This bias is a complex function of the noise variance and the m non-null eigenvalues of the loading matrix $\mathbf{\Lambda\Lambda}'$. The above CLT is still valid if $\tilde{c}_n=(p-m)/n$ is substituted for c. Now if indeed $p\ll n$, i.e. the dimension p is infinitely smaller than the sample size n, so that $\tilde{c}_n\simeq 0$ and $b(\sigma^2)\simeq 0$, hence

$$\frac{p-m}{\sigma^2\surd(2c)}(\hat{\sigma}^2-\sigma^2)+b(\sigma^2)\simeq\frac{\surd(p-m)}{\sigma^2\surd 2}\surd n(\hat{\sigma}^2-\sigma^2)\xrightarrow{\mathcal{D}}\mathcal{N}(0,1).$$

This is the CLT (4) for $\hat{\sigma}^2$ known under the classical low dimensional scheme. In a sense, theorem 1 constitutes a natural and continuous extension of the classical CLT to the high dimensional context.

Theorem 1 recommends correction of the negative bias of $\hat{\sigma}^2$. As the bias depends on σ^2 which we want to estimate, a natural correction is to use the plug-in estimator

$$\hat{\sigma}^2_*=\hat{\sigma}^2+\frac{b(\hat{\sigma}^2)}{p-m}\hat{\sigma}^2\surd(2c_n). \tag{7}$$

This estimator will be hereafter referred as the *bias-corrected estimator*. The following CLT is a direct consequence of theorem 1.

Theorem 2. We assume the same conditions as in theorem 1. Then, we have

$$\frac{p-m}{\sigma^2\surd(2c_n)}(\hat{\sigma}^2_*-\sigma^2)\xrightarrow{\mathcal{D}}\mathcal{N}(0,1).$$

The proof is given in Appendix A. Compared with the MLE $\hat{\sigma}^2$ in theorem 1, the bias-corrected estimator $\hat{\sigma}^2_*$ has no longer a bias after normalization by $(p-m)/\{\sigma^2\surd(2c_n)\}$, and it should be a much better estimator than $\hat{\sigma}^2$.

For the implementation of $\hat{\sigma}^2_*$ in practice, we need the value of $b(\hat{\sigma}^2)$ which depends on the unknown spike values $\{\alpha_i\}$. We remark that only a consistent estimate of $b(\hat{\sigma}^2)$ is needed here and this is achieved by substituting some consistent estimates $\hat{\alpha}_i$ for α_i in $b(\hat{\sigma}^2)$. This is done as follows. Following theorem 1, $\hat{\sigma}^2\to^{\mathrm{P}}\sigma^2$. Then using the function ψ in result (6) and, by solving in α_i the equation $\lambda_{n,i}=\psi(\hat{\sigma}^2,p/n,\alpha_i)$, we find an estimator $\hat{\alpha}_i$ for α_i. Since $p/n\to c$, $\hat{\sigma}^2\to^{\mathrm{P}}\sigma^2$ and ψ is known to be invertible, we deduce easily that $\hat{\alpha}_i\to^{\mathrm{P}}\alpha_i$. This procedure will be used for real data analysis in Section 2.4.

2.1. Monte Carlo experiments

We first check by simulation the effect of bias correction obtained in $\hat{\sigma}^2_*$ and its asymptotic normality. Independent Gaussian samples of size n are considered in three different settings: model 1,

$$\operatorname{spec}(\mathbf{\Sigma})=(25,16,9,0,\ldots,0)+\sigma^2(1,\ldots,1),\sigma^2=4,c=1;$$

model 2,

$$\operatorname{spec}(\mathbf{\Sigma})=(4,3,0,\ldots,0)+\sigma^2(1,\ldots,1),\sigma^2=2,c=0.2;$$

model 3,

$$\operatorname{spec}(\mathbf{\Sigma})=(12,10,8,8,0,\ldots,0)+\sigma^2(1,\ldots,1),\sigma^2=3,c=1.5.$$

Table 1. Comparison between the empirical and the theoretical bias

Model	*p*	*n*	*Empirical bias*	*Theoretical bias*	*|Difference|*
1	100	100	−0.1556	−0.1589	0.0024
	400	400	−0.0391	−0.0388	0.0003
	800	800	−0.0197	−0.0193	0.0003
2	20	100	−0.0625	−0.0704	0.0052
	80	400	−0.0166	−0.0162	0.0027
	200	1000	−0.0064	−0.0064	0.0011
3	150	100	−0.1609	−0.1634	0.0025
	600	400	−0.0401	−0.0400	0.0001
	1500	1000	−0.0161	−0.0159	0.0002

In Table 1, we compare the empirical bias of $\hat{\sigma}^2$ (i.e. the empirical mean of $\hat{\sigma}^2 - \sigma^2 = \{1/(p-m)\}\Sigma_{i=m+1}^{p}\lambda_{n,i} - \sigma^2$) over 1000 replications with the theoretical bias $-\sigma^2\surd(2c) \times b(\sigma^2)/(p-m)$. In all the three models, the empirical and theoretical biases are close to each other. As expected, their difference vanishes when p and n increase. Table 1 also shows that this bias is quite significant even for large dimension and sample size such as $(p,n)=(1500,1000)$. In addition, we have drawn the histograms from 1000 replications of $(p-m)\{\sigma^2\surd(2c_n)\}^{-1}(\hat{\sigma}^2 - \sigma^2) + b(\sigma^2)$ of the three models above, with sample size $n=100$ and dimensions $p=cn$ and they match very well the density of the standard Gaussian distribution (see the on-line supplementary report).

Next, we compare our bias-corrected estimator $\hat{\sigma}^2_*$ with the MLE $\hat{\sigma}^2$ and other three existing estimators in the literature. For the reader's convenience, we recall their definitions.

(a) The estimator $\hat{\sigma}^2_{\mathrm{KN}}$ of Kritchman and Nadler (2008) is defined as the solution of the following non-linear system of $m+1$ equations involving the $m+1$ unknowns $\hat{\rho}_1,\ldots,\hat{\rho}_m$ and $\hat{\sigma}^2_{\mathrm{KN}}$:

$$\hat{\sigma}^2_{\mathrm{KN}} - \frac{1}{p-m}\left\{\sum_{j=m+1}^{p}\lambda_{n,j} + \sum_{j=1}^{m}(\lambda_{n,j}-\hat{\rho}_j)\right\} = 0,$$

and

$$\hat{\rho}_j^2 - \hat{\rho}_j\left(\lambda_{n,j} + \hat{\sigma}^2_{\mathrm{KN}} - \hat{\sigma}^2_{\mathrm{KN}}\frac{p-m}{n}\right) + \lambda_{n,j}\hat{\sigma}^2_{\mathrm{KN}} = 0, \qquad j=1,\ldots,m.$$

We use the code that is available from the authors' Web page to carry out the simulation. Note that $\hat{\sigma}^2_{\mathrm{KN}}$ is only implicitly defined and no precise asymptotic analysis was provided in Kritchman and Nadler (2008) for $\hat{\sigma}^2_{\mathrm{KN}}$. We mention one common feature that is shared by $\hat{\sigma}^2_{\mathrm{KN}}$ and $\hat{\sigma}^2_*$: both methods use the same relationship between the population spike eigenvalues and their asymptotic limits which are equation (6) in our paper and equation (16) in Kritchman and Nadler (2008) (this leads to the second equation in the system above satisfied by $\hat{\sigma}^2_{\mathrm{KN}}$).

(b) The estimator $\hat{\sigma}^2_{\mathrm{US}}$ of Ulfarsson and Solo (2008) is defined as the ratio

$$\hat{\sigma}^2_{\mathrm{US}} = \frac{\mathrm{median}(\lambda_{n,m+1},\ldots,\lambda_{n,p})}{m_{p/n,1}},$$

where $m_{\alpha,1}$ is the median of the Marčenko–Pastur distribution $F_{\alpha,1}$.

Table 2. Comparison between four existing estimators and the proposed $\hat{\sigma}^2_*$ in terms of ratios of MSEs: $\text{MSE}(\hat{\sigma}^2)/\text{MSE}(\hat{\sigma}^2_*)$, $\text{MSE}(\hat{\sigma}^2_{\text{KN}})/\text{MSE}(\hat{\sigma}^2_*)$, $\text{MSE}(\hat{\sigma}^2_{\text{US}})/\text{MSE}(\hat{\sigma}^2_*)$ and $\text{MSE}(\hat{\sigma}^2_{\text{median}})/\text{MSE}(\hat{\sigma}^2_*)$

Model	p	n	σ^2	$\hat{\sigma}^2$	$\hat{\sigma}^2_{\text{KN}}$	$\hat{\sigma}^2_{\text{US}}$	$\hat{\sigma}^2_{\text{median}}$
1	100	100	4	7.8232	1.0130	14.6394	1.5085
	400	400		8.5905	0.9980	25.5941	1.6429
	800	800		8.1162	1.0019	39.9444	1.6639
2	20	100	2	1.7045	1.0220	2.4980	1.5926
	80	400		2.0406	1.0045	3.8686	1.5433
	200	1000		1.9729	1.0011	3.8731	1.5427
3	150	100	3	19.2114	1.2292	41.7319	1.4274
	600	400		20.8471	0.9958	48.3130	1.6096
	1500	1000		21.6207	1.0001	51.9302	1.8071

(c) The estimator $\hat{\sigma}^2_{\text{median}}$ of Johnstone and Lu (2009) is defined as the median of the p sample variances (the data $\{x_{ij}\}$ are assumed centred):

$$\hat{\sigma}^2_{\text{median}} = \text{median}\left(\frac{1}{n}\sum_{i=1}^{n} x^2_{ij}, \quad 1 \leqslant j \leqslant p\right).$$

Table 2 presents the ratios of the empirical mean-squared errors MSE of these estimators over the empirical MSE of the bias-corrected estimator $\hat{\sigma}^2_*$. The performances of $\hat{\sigma}^2_*$ and $\hat{\sigma}^2_{\text{KN}}$ are similar but $\hat{\sigma}^2_*$ is slightly better. The estimator $\hat{\sigma}^2_{\text{median}}$ is better than $\hat{\sigma}^2_{\text{US}}$ and the MLE $\hat{\sigma}^2$. But $\hat{\sigma}^2_{\text{median}}$ and $\hat{\sigma}^2_{\text{US}}$ perform poorly compared with $\hat{\sigma}^2_*$ and $\hat{\sigma}^2_{\text{KN}}$. The reader is, however, reminded that the theoretic properties of $\hat{\sigma}^2_{\text{KN}}$, $\hat{\sigma}^2_{\text{US}}$ and $\hat{\sigma}^2_{\text{median}}$ are unknown and so far they have been checked via simulations only. A careful look at the defining formula of both $\hat{\sigma}^2_{\text{US}}$ and $\hat{\sigma}^2_{\text{median}}$ reveals that these estimators are close to $\hat{\sigma}^2$, all of them being close to the average or median of sample eigenvalues. Therefore they might have a similar performance $\hat{\sigma}^2$.

2.2. Extension to non-Gaussian data

In this section, we provide some extensions of the main results to cover non-Gaussian data. Following a common approach in high dimensional statistics (Bai and Saranadasa, 1996), we assume that $\mathbf{x}_i$ can be generated as

$$\mathbf{x}_i = \mathbf{A}\mathbf{y}_i + \boldsymbol{\mu}, \tag{8}$$

where $\mathbf{A} = \boldsymbol{\Sigma}^{1/2}$ and $\mathbf{y}_i = \{y_{ij}\}_{1\leqslant j\leqslant p}$ has p independently and identically distributed and standardized components. We set $\gamma = E|y_{11}|^4 - 1$ ($\gamma = 2$ under the normal assumption).

Theorem 3. Consider the PPCA model (1) where the observation vectors $\{\mathbf{x}_i\}_{1\leqslant i\leqslant n}$ are generated as in expression (8). Assume that $p \to \infty, n \to \infty$ and $c_n = p/(n-1) \to c > 0$. Then, we have

$$\frac{p-m}{\sigma^2\sqrt{(\gamma c)}}(\hat{\sigma}^2 - \sigma^2) + b(\sigma^2) \xrightarrow{\mathcal{D}} \mathcal{N}(0,1).$$

The proof is given in Appendix A. Similarly to the plug-in estimator $\hat{\sigma}^2_*$ in expression (7) for Gaussian data, we define the new plug-in estimator

$$\hat{\sigma}^2_{*0} = \hat{\sigma}^2 + \frac{b(\hat{\sigma}^2)}{p-m}\hat{\sigma}^2\sqrt{(\gamma c_n)}. \tag{9}$$

When $\gamma=2$ (the Gaussian case), $\hat{\sigma}^2_{*0}$ coincides with $\hat{\sigma}^2_{*}$.

Theorem 4. We assume the same conditions as in theorem 3. Then, we have

$$\frac{p-m}{\sigma^2\sqrt{(\gamma c_n)}}(\hat{\sigma}^2_{*0}-\sigma^2)\xrightarrow{\mathcal{D}}\mathcal{N}(0,1).$$

The proof of theorem 4 is omitted because it is the same as for theorem 2. For the implementation of $\hat{\sigma}^2_{*0}$, there is, however, a new problem to solve with non-Gaussian data, namely the parameter γ in equation (9) needs to be estimated first. Again we use random-matrix theory to resolve this issue.

Proposition 1. Under the same conditions as in theorem 3, as $p,n\rightarrow\infty$,

$$\sum_{i=1}^{p}\lambda^2_{n,i}-p(\beta_2+c_n\beta_1^2)\xrightarrow{\mathcal{D}}\mathcal{N}\{c\sigma^4(\gamma-1),v\},$$

where $v>0$ denotes a (computable) asymptotic variance, and

$$\beta_1=\sigma^2+\frac{1}{p}\sum_{j=1}^{m}\alpha_j,$$

$$\beta_2=\sigma^4+\frac{1}{p}\sum_{j=1}^{m}\alpha_j^2+\frac{2}{p}\sigma^2\sum_{j=1}^{m}\alpha_j.$$

This result does not provide directly a consistent estimator of γ. Therefore, the bootstrap method is applied to the sample eigenvalues $\{\lambda_{n,i}\}_{1\leqslant i\leqslant p}$ to obtain say B bootstrapped samples $\{\lambda^*_{n,i}\}_{1\leqslant i\leqslant p}$. We find the bootstrap mean w^* of $w=\{\Sigma_{i=1}^{p}\lambda^2_{n,i}-p(\beta_2+c_n\beta_1^2)\}/(c\sigma^4)$ (here σ^2 is approximated by the MLE $\hat{\sigma}^2$), and finally by letting $w^*=\hat{\gamma}^*-1$ we find a bootstrap estimator $\hat{\gamma}^*$ of the unknown γ. Plugging $\hat{\gamma}^*$ into equation (9), the final bias-corrected estimator that we propose is

$$\hat{\sigma}^2_{**}=\hat{\sigma}^2+\frac{b(\hat{\sigma}^2)}{p-m}\hat{\sigma}^2\sqrt{(\hat{\gamma}^*c_n)}. \tag{10}$$

We conclude the section by some simulation experiments to check the performance of $\hat{\sigma}^2_{**}$ for non-Gaussian data. We start with the same setting of the covariance matrix of Table 1 and use independent gamma and continuous uniform-distributed variables for the components of the $\mathbf{x}_i$s. The gamma-distributed data are drawn from gamma(k,θ) with fixed shape parameter $k=2$ and accordingly determined scale parameter θ; the uniform-distributed data are drawn from $U(a,b)$ with fixed $a=0$ and accordingly determined b. The bootstrap samples are repeated $B=500$ times. The simulation results are summarized in Table 3. The estimator $\hat{\sigma}^2_{**}$ has a very good performance for both tested non-Gaussian distributions. The median absolute difference MAD and MSE decrease when p and n increase in all model settings.

2.3. Determination of the number of principal components

So far we have assumed that the number m of PCs is known. It is, however, desirable to estimate m directly from the data. In the literature, consistent estimators of m have been proposed in the high dimensional context, e.g. in Kritchman and Nadler (2008), Ulfarsson and Solo (2008), Onatski (2009) and Passemier and Yao (2012). As a benchmark work, Bai and Ng (2002) proposed six criteria to determine the number of PCs (or factors) under the framework of large cross-sections N and large time dimensions T. These criteria are popular and widely used in the factor modelling literature: for example, they are one of the starting blocks of the newly

Table 3. Empirical mean, MAD and MSE of $\hat{\sigma}^2_{**}$ for gamma and uniform samples

Model	*p*	*n*	σ^2	*Results for gamma distribution*			*Results for continuous uniform distribution*		
				$\hat{\sigma}^2_{**}$	*MAD*	*MSE*	$\hat{\sigma}^2_{**}$	*MAD*	*MSE*
1	100	100	4	4.0807	0.1697	0.0411	4.0150	0.1052	0.0161
	400	400		4.0561	0.0571	0.0040	4.0424	0.0429	0.0022
	800	800		4.0304	0.0306	0.0011	4.0236	0.0237	0.0006
2	20	100	2	1.9570	0.1150	0.0194	1.9175	0.0849	0.0089
	80	400		1.9978	0.0241	0.0009	1.9858	0.0166	0.0004
	200	1000		2.0016	0.0090	0.0001	1.9968	0.0054	$<10^{-4}$
3	150	100	3	3.3637	0.3637	0.1415	3.3336	0.3336	0.1143
	600	400		3.1060	0.1060	0.0116	3.0957	0.0957	0.0093
	1500	1000		3.0434	0.0434	0.0019	3.0391	0.0391	0.0015

proposed estimator POET in Fan *et al.* (2013). Note that the dimension–sample size pair is denoted here as (N,T) instead of (p,n). Three of these criteria that are applicable to PPCA models are

$$\mathrm{PC}_j(m) = V(m, \hat{F}^m) + m\hat{\sigma}^2_{\mathrm{BN}}\, g_j(N,T), \qquad j \in \{1,2,3\}, \tag{11}$$

where $\hat{\sigma}^2_{\mathrm{BN}}$ is a consistent estimate of $(NT)^{-1}\Sigma_{i=1}^N \Sigma_{j=1}^T E(e_{ij})^2$, $V(m,\hat{F}^m) = (NT)^{-1}\Sigma_{i=1}^N \hat{\mathbf{e}}_i'\hat{\mathbf{e}}_i$ and $g_j(N,T)$ denote the penalty functions

$$g_1(N,T) = \frac{N+T}{NT}\ln\left(\frac{NT}{N+T}\right),$$

$$g_2(N,T) = \frac{N+T}{NT}\ln(\tilde{N}),$$

$$g_3(N,T) = \frac{\ln(\tilde{N})}{\tilde{N}},$$

with $\tilde{N} = \min\{N,T\}$. The corresponding estimators of the number of PCs are $\hat{m}_j = \arg\min_{0\leqslant m\leqslant m_0} \mathrm{PC}_j(m)$, $j \in \{1,2,3\}$, where m_0 is a predetermined maximum value of m. In applications, $\hat{\sigma}^2_{\mathrm{BN}}$ is replaced by $V(m_0, \hat{F}^{m_0})$. The calculations of $\hat{\sigma}^2_{\mathrm{BN}}$ and $V(m,\hat{F}^m)$ have no explicit formula and are based on the estimation of the residuals $\{\hat{\mathbf{e}}_i\}$. It is worth mentioning that $V(m,\hat{F}^m)$ and $\hat{\sigma}^2_{\mathrm{BN}}$ are indeed the estimates of the noise variance if the underlying model is the PPCA model.

To start with and to assess the quality of our bias-corrected estimator $\hat{\sigma}^2_*$ in the current context, we substitute $\hat{\sigma}^2_*$ for empirical $V(m,\hat{F}^m)$ and $\hat{\sigma}^2_{\mathrm{BN}}$ in the criteria PC_js. Note that the noise variance estimator $\hat{\sigma}^2_*$ depends on the supposed number m of PCs, and we let $\hat{\sigma}^2_*(m) = \hat{\sigma}^2_*$ denote explicitly this dependence. The modified criteria and estimators using $\hat{\sigma}^2_*(m)$ are thus

$$\mathrm{PC}_j^*(m) = \hat{\sigma}^2_*(m) + m\,\hat{\sigma}^2_*(m_0)\,g_j(N,T),$$

and

$$\hat{m}_j^* = \arg\min_{0\leqslant m\leqslant m_0} \mathrm{PC}_j^*(m), \qquad j \in \{1,2,3\}, \tag{12}$$

respectively. These modified criteria PC_j^*s will be compared below with their original counterparts by simulation.

Under appropriate conditions, Bai and Ng (2002) established the consistency of the criteria PC_j when both N and T grow to ∞. A careful examination of their method reveals that the modified criteria PC_j^* are also consistent under the same conditions (this thus means that potential differences between the two families of criteria are of higher asymptotic order). There is, however, a main issue here: such consistency results require that the variances of the PCs (or their strengths) grow to ∞ with the dimension N; see assumption B of Bai and Ng (2002). This *pervasiveness* assumption is not satisfied in our context where these variances can be weaker and remain bounded. Consequently, the proof of Bai and Ng (2002) does not apply here and we are forced to seek a new asymptotic result. We thus introduce the new penalty function

$$g(N,T)=\frac{(c+2\surd c)(1+T/N^{1+\delta})}{N} \tag{13}$$

and define the new criterion

$$\mathrm{PC}^*(m)=\hat{\sigma}_*^2(m)+m\,\hat{\sigma}_*^2(m_0)\,g(N,T). \tag{14}$$

Here $\delta>0$ is a small prefixed constant. As another main contribution of the paper, we establish the consistency of the corresponding estimator

$$\hat{m}^*=\arg\min_{0\leqslant m\leqslant m_0}\mathrm{PC}^*(m). \tag{15}$$

Theorem 5. We assume the same conditions as in theorem 1: in particular $N,T\rightarrow\infty$ and $c_n=N/(T-1)\rightarrow c>0$. With the condition $\alpha_i>\sigma^2\surd c$, we have $\lim_{N,T\rightarrow\infty}\mathrm{Prob}(\hat{m}^*=\bar{m})=1$ where $\bar{m}$ is the true number of PCs.

The proof is given in Appendix A. Simulation experiments are conducted to show the performance of the new estimator $\hat{m}^*$ in comparison with both the modified estimators $\hat{m}_j^*$ and the original $\hat{m}_j$s. As in Bai and Ng (2002), the data are generated from the model

$$X_{it}=\sum_{j=1}^{m}\lambda_{ij}F_{tj}+\surd\theta e_{it},$$

where the PCs, the loadings and the errors e_{it} are $\mathcal{N}(0,1)$ variates, the common component of X_{it} has variance m and the idiosyncratic component has variance θ. The noise variance is $\sigma^2=\theta$ and $\mathbf{\Lambda}=(\lambda_{ij})$. Typically, a PC corresponding to α_j is detectable when $\alpha_j\geqslant\surd(N/T)\theta$; see expression (6). We conduct extensive simulations by reproducing the configuration of N and T that was used in Bai and Ng (2002). In all the experiments, the same value of $\delta=0.05$ is used in equation (13).

Tables 4 and 5 report the empirical means of the estimator of the number of PCs over 1000 replications, for $m=1$ and $m=5$ respectively, with standard errors in parentheses. When a standard error is actually 0, no standard error is thus indicated. For all cases, the predetermined maximum number m_0 of PCs is set to 8. When the true number of PCs is 1 (Table 4), the new criterion PC^* and the modified criteria PC_j^* can correctly detect the number almost surely and the corresponding standard errors are all 0s. In comparison, there are 11 cases where the original criteria PC_j lose efficiency in finding the true number of PCs with a non-zero standard error. In the small dimensions situations (the last five rows), all the modified PC_j^* and the original PC_j fail when the value of N is 10: they all report the maximum value m_0. But the new criterion PC^* outperforms the others in all cases in terms of mean and standard error. Meanwhile,

Table 4. Comparison between PC^*, PC_j^* and PC_j for $m = 1$ and $\theta = 1$

N	T	PC^*	PC_1^*	PC_2^*	PC_3^*	PC_1	PC_2	PC_3
100	40	1.00	1.00	1.00	1.00	1.17 (0.37)	1.01 (0.10)	3.78 (0.75)
100	60	1.00	1.00	1.00	1.00	1.00	1.00	3.63 (0.76)
200	60	1.00	1.00	1.00	1.00	1.00	1.00	1.00
500	60	1.00	1.00	1.00	1.00	1.00	1.00	1.00
1000	60	1.00	1.00	1.00	1.00	1.00	1.00	1.00
2000	60	1.00	1.00	1.00	1.00	1.00	1.00	1.00
100	100	1.00	1.00	1.00	1.00	1.00	1.00	5.36 (0.80)
40	100	1.00	1.00	1.00	1.00	1.79 (0.72)	1.19 (0.40)	4.91 (0.90)
60	100	1.00	1.00	1.00	1.00	1.01 (0.08)	1.00	4.30 (0.85)
60	200	1.00	1.00	1.00	1.00	1.00	1.00	1.02 (0.16)
10	50	7.19 (1.92)	8.00	8.00	8.00	8.00	8.00	8.00
10	100	4.95 (3.13)	8.00	8.00	8.00	8.00	8.00	8.00
20	100	1.00 (0.03)	1.01 (0.15)	1.01 (0.12)	1.08 (0.53)	6.96 (0.88)	6.35 (0.98)	7.84 (0.40)
100	10	2.10 (2.52)	1.08 (0.73)	1.03 (0.50)	1.15 (1.01)	8.00	8.00	8.00
100	20	1.00	1.00 (0.03)	1.00 (0.03)	1.00 (0.03)	5.88 (0.76)	5.12 (0.77)	7.35 (0.63)

Table 5. Comparison between PC^*, PC_j^* and PC_j for $m = 5$ and $\theta = 3$

N	T	PC^*	PC_1^*	PC_2^*	PC_3^*	PC_1	PC_2	PC_3
100	40	5.00	4.91 (0.30)	4.81 (0.41)	4.99 (0.11)	5.00 (0.03)	5.00	5.59 (0.57)
100	60	5.00	5.00 (0.04)	4.99 (0.11)	5.00	5.00	5.00	5.58 (0.57)
200	60	5.00	5.00	5.00	5.00	5.00	5.00	5.00
500	60	5.00	5.00	5.00	5.00	5.00	5.00	5.00
1000	60	5.00	5.00	5.00	5.00	5.00	5.00	5.00
2000	60	5.00	5.00	5.00	5.00	5.00	5.00	5.00
100	100	5.00	5.00	5.00	5.00	5.00	5.00	6.84 (0.65)
40	100	4.98 (0.14)	4.97 (0.17)	4.92 (0.27)	5.00 (0.04)	5.02 (0.12)	5.00	6.22 (0.66)
60	100	5.00	5.00 (0.04)	4.99 (0.08)	5.00	5.00	5.00	6.03 (0.64)
60	200	5.00	5.00	5.00	5.00	5.00	5.00	6.03 (0.03)
10	50	7.47 (1.00)	8.00	8.00	8.00	8.00	8.00	8.00
10	100	5.77 (1.53)	8.00	8.00	8.00	8.00	8.00	8.00
20	100	3.74 (0.84)	4.74 (0.51)	4.62 (0.57)	4.92 (0.45)	7.11 (0.63)	6.65 (0.64)	7.85 (0.37)
100	10	6.66 (1.97)	4.59 (1.99)	4.35 (1.91)	4.88 (2.09)	8.00	8.00	8.00
100	20	4.68 (0.51)	3.86 (0.79)	3.69 (0.81)	4.13 (0.73)	6.74 (0.63)	6.19 (0.62)	7.77 (0.43)

the modified criteria PC_j^* globally perform better than the original PC_js and this establishes the superiority of the bias-corrected estimator $\hat{\sigma}_*^2$. In Table 5, the common component has variance 5 and the idiosyncratic component has a smaller variance 3, and the situation is a little more difficult. We can, however, draw the same conclusion that the new criterion PC^* outperforms the other criteria in all cases tested, and the modified criteria PC_j^* have an overall better performance than the original PC_js. In both Table 4 and Table 5, only a part of the tested combinations of N and T is reported and the other combinations where all criteria detect the right number m with zero error have been omitted. Additional simulation results and tables are in the on-line supplementary report. In conclusion, the proposed criterion has the best performance in determining the number of PCs, and the modified criteria perform better by using the bias-corrected estimator proposed in this paper for the PPCA model.

Table 6. Comparison between the modified and the original criteria

Data set	PC^*	PC_1^*	PC_2^*	PC_3^*	PC_1	PC_2	PC_3
1 ($m_0=15$)	2	2	2	2	4	4	6
2 ($m_0=20$)	13	18	18	19	20	20	20

2.4. Real data

Though the new and modified estimators seem to perform better than the original estimators in the simulation experiments, we now compare them on two real data sets. The first data set contains stock returns. Following Bai and Ng (2002), we extract data from the Center for Research in Security Prices US stock database by using the monthly returns for all common stocks listed in the New York Stock Exchange, Amex and Nasdaq over 20 years (from January 1991 to December 2010). Stocks that do not trade for a cumulative 2 years during the period have been deleted. The final data set includes 1913 stocks with 240 monthly returns for each of them ($T=240; N=1913$). The data set does not match exactly the set that was used in Bai and Ng (2002) as they selected 4883 firms for the shorter period from January 1994 to December 1998; however, this selected data set is not publicly available. The second data set is the functional magnetic resonance imaging data set. This data set is freely available from the Web site `http://afni.nimh.nih.gov/afni/`. A human brain was scanned when the person performed finger–thumb opposition. There are $T=124$ observations on 21 brain slices. We pick out one brain slice and keep only the variables (pixels) that significantly corresponded to brain tissue, so that $N=1126$ variables are selected. We transform both data sets so that each series has mean 0. The results of rank estimates of the new and modified criteria on these two data sets are shown in Table 6. The original criteria PC_j display significant variation for the first data set and fail for the second by only reporting the maximum value $m_0=20$. In contrast, the new criterion PC^* and the modified criteria PC_j^* with the proposed variance estimator $\hat{\sigma}_*^2$ seem mutually consistent by giving very close if not identical rank estimates. In particular, the original criteria PC_j have a significant overestimate effect and this is much reduced and stabilized either with a more accurate estimate of the noise variance by $\hat{\sigma}_*^2$ or with the new penalty function $g(N,T)$ in equation (13).

3. Application to the goodness-of-fit test of a probabilistic principal component analysis model

As an additional application of the bias-corrected estimator σ_*^2, we consider the following goodness-of-fit test for the PPCA model (1). The null hypothesis is

$$\mathcal{H}_0: \mathbf{\Sigma}=\mathbf{\Lambda}\mathbf{\Lambda}'+\sigma^2\mathbf{I}_p,$$

where the number of PCs m is specified. Following Anderson and Rubin (1956), the LRT statistic is

$$T_n=-nL^*, \qquad L^*=\sum_{j=m+1}^{p}\log\left(\frac{\lambda_{n,j}}{\hat{\sigma}^2}\right),$$

and $\hat{\sigma}^2$ is the MLE (3) of the variance. Keeping p fixed while letting $n\to\infty$, classical low dimensional theory states that T_n converges to χ_q^2, where $q=p(p+1)/2+m(m-1)/2-pm-1$.

Table 7. Comparison of the empirical size of the LRT and CLRT in various settings

Model	*p*	*n*	*Empirical size of CLRT*	*Empirical size of LRT*
1	90	100	0.0522	0.9997
	180	200	0.0515	1.0000
	720	800	0.0483	1.0000
2	20	100	0.0375	0.0321
	80	400	0.0440	0.0368
	200	1000	0.0481	0.0514
4	5	500	0.0122	0.0475
	10	500	0.0217	0.0482
	50	500	0.0421	0.0419
	100	500	0.0438	0.0424
	200	500	0.0498	0.2216
	250	500	0.0501	0.7416
	300	500	0.0461	0.9991

However, this classical approximation is again useless in large dimensional situations. Indeed, this criterion leads to a high false positive rate (Table 7).

In a way similar to Section 2, we now construct a corrected version of T_n by using the calculus that was done in Bai *et al.* (2009) and Zheng (2012). As we consider the logarithm of the eigenvalues of the sample covariance matrix, we shall assume in what follows that $p<n$ and $c<1$ to avoid null eigenvalues.

Theorem 6. Assume the same conditions as in theorem 1 and in addition $c<1$. Then, we have

$$v(c)^{-1/2}\{L^*-m(c)-p\,h(c_n)+\eta+(p-m)\log(\beta)\}\xrightarrow{\mathcal{D}}\mathcal{N}(0,1),$$

where

$$m(c)=\frac{\log(1-c)}{2},$$

$$h(c_n)=\frac{c_n-1}{c_n}\log(1-c_n)-1,$$

$$\eta=\sum_{i=1}^{m}\log(1+c\sigma^2\alpha_i^{-1}),$$

$$\beta=1-\frac{c}{p-m}\left(m+\sigma^2\sum_{i=1}^{m}\alpha_i^{-1}\right),$$

$$v(c)=-2\log(1-c)+\frac{2c}{\beta}\left(\frac{1}{\beta}-2\right).$$

The proof is given in the on-line supplementary report. The above statistic depends on the unknown variance σ^2 and the spike eigenvalues $\{\alpha_i\}$. First of all, as explained in Section 2, consistent estimates of $\{\alpha_i\}$ are available. By using these estimates and substituting bias-corrected estimate $\hat{\sigma}^2_*$ for σ^2, we obtain consistent estimates $\hat{v}(c_n)$, $\hat{\eta}$ and $\hat{\beta}$ of $v(c)$, η and β respectively. Therefore, to test $\mathcal{H}_0$, it is natural to use the statistic

$$\Delta_n:=\hat{v}(c_n)^{-1/2}\{L^*-m(c_n)-p\,h(c_n)+\hat{\eta}+(p-m)\log(\hat{\beta})\}.$$

Since Δ_n is asymptotically standard normal, the critical region $\{\Delta_n > q_\alpha\}$ where q_α is the αth upper quantile of the standard normal distribution, will have an asymptotic size α. This test is referred to as the CLRT.

Next, we present some simulation experiments to compare the classical LRT and the CLRT. We consider again models 1 and 2 that were described in Section 2, and a new model (model 4): model 1,

$$\text{spec}(\mathbf{\Sigma}) = (25, 16, 9, 0, \ldots, 0) + \sigma^2(1, \ldots, 1), \sigma^2 = 4, c = 0.9;$$

model 2,

$$\text{spec}(\mathbf{\Sigma}) = (4, 3, 0, \ldots, 0) + \sigma^2(1, \ldots, 1), \sigma^2 = 2, c = 0.2;$$

model 4,

$$\text{spec}(\mathbf{\Sigma}) = (8, 7, 0, \ldots, 0) + \sigma^2(1, \ldots, 1), \sigma^2 = 1, \text{ varying } c.$$

Table 7 presents the empirical sizes of the LRT and the CLRT. For the LRT, we use the correction that was proposed by Bartlett (1950), i.e. replacing $T_n = -nL^*$ by $\tilde{T}_n = -\{n - (2p+11)/6 - 2m/3\}L^*$. The computations are done under 10000 independent replications and the nominal test level is 0.05. The empirical sizes of the CLRT are very close to the nominal size, except when the ratio p/n is very small (less than 0.1). In contrast, the empirical sizes of the classical LRT are much higher than the nominal level especially when c is not too small, and the test will always reject the null hypothesis when p becomes large. In particular, when $p/n \geqslant \frac{1}{2}$, the LRT tends to reject automatically the null hypothesis.

4. Conclusions

In this paper, we propose a bias-corrected estimator of the noise variance for PPCA models in the high dimensional framework. The main appeal of our estimator is that it is developed under the assumption that $p/n \to c > 0$ as $p, n \to \infty$ and is thus appropriate for a wide range of large dimensional data sets. Extensive Monte Carlo experiments demonstrated the superiority of the proposed estimator over several existing estimators (however, no theoretical justification has been proposed in the literature for these estimators). In addition, by implementing the proposed estimator of the noise variance within the well-known determination algorithms for the number of PCs that was proposed by Bai and Ng (2002), we construct a new joint consistent estimator of the pair (m, σ^2) with a new penalty function to cope with non-pervasive PCs. In an additional application of our methodology, we develop asymptotic theory of the goodness-of-fit test for high dimensional PPCA models. The overall message from the paper is that in a high dimensional PPCA model, when an estimator of the noise variance σ^2 is needed, the bias-corrected estimator $\hat{\sigma}^2_*$ from the paper should be recommended.

To conclude, we mention an important question that requires further investigation, namely the effect of the size of m and of the PC eigenvalues on the methodology that is developed in this paper. In the numerical simulations, m is typically small compared with $\min\{p, n\}$. Can we expect different behaviour if

(a) some of the PC eigenvalues are fairly big compared with σ^2, say of the order $O(p)$, as is often the case in some econometric problems (Fan *et al.*, 2008) or
(b) when m is relatively big, e.g. increasing with p, whereas many of the PC eigenvalues are small, possibly below the size where the phase transition of eigenvalues take place?

Unfortunately, these questions go much beyond the scope the existing literature including this paper: for instance we are not aware of any work that is capable of integrating both large and small PC eigenvalues. Similarly, the treatment of varying numbers m of PC components will require the development of new mathematical techniques. Needless to say, these questions are of fundamental importance and are worth much research effort in the future.

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Appendix A

A.1. Proof of theorem 1

We have $(p-m)\hat{\sigma}^2 = \Sigma_{i=1}^{p} \lambda_{n,i} - \Sigma_{i=1}^{m} \lambda_{n,i}$. By expression (6),

$$\sum_{i=1}^{m} \lambda_{n,i} \to \sum_{i=1}^{m} \left(\alpha_i + \frac{c\sigma^4}{\alpha_i} \right) + \sigma^2 m(1+c) \qquad \text{almost surely.} \tag{16}$$

For the first term, we have

$$\begin{aligned} \sum_{i=1}^{p} \lambda_i &= p \int x \, \mathrm{d}F_n(x) \\ &= p \int x \, \mathrm{d}(F_n - F_{c_n,H_n})(x) + p \int x \, \mathrm{d}F_{c_n,H_n}(x) \\ &= G_n(x) + p \int x \, \mathrm{d}F_{c_n,H_n}(x). \end{aligned}$$

By proposition 1 in the on-line supplement report, the first term is asymptotically normal

$$G_n(x) = \sum_{i=1}^{p} \lambda_{n,i} - p \int x \, \mathrm{d}F_{c_n,H_n}(x) \xrightarrow{\mathcal{D}} \mathcal{N}\{m(x), v(x)\},$$

with asymptotic mean

$$m(x) = 0 \tag{17}$$

and asymptotic variance

$$v(x) = 2c\sigma^4. \tag{18}$$

Furthermore, by lemma 1 of Bai *et al.* (2010),

$$\int x \, \mathrm{d}F_{c_n,H_n}(x) = \int t \, \mathrm{d}H_n(t) = \sigma^2 + \frac{1}{p} \sum_{i=1}^{m} \alpha_i.$$

So we have

$$\sum_{i=1}^{p} \lambda_{n,i} - p\sigma^2 - \sum_{i=1}^{m} \alpha_i \xrightarrow{\mathcal{D}} \mathcal{N}(0, 2c\sigma^4). \tag{19}$$

By results (16) and (19) and using Slutsky's lemma, we obtain

$$(p-m)(\hat{\sigma}^2 - \sigma^2) + c\sigma^2 \left(m + \sigma^2 \sum_{i=1}^{m} \frac{1}{\alpha_i} \right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 2c\sigma^4).$$

A.2. Proof of theorem 2

We have

$$\begin{aligned} \frac{p-m}{\sigma^2 \surd(2c_n)} (\hat{\sigma}_*^2 - \sigma^2) &= \frac{p-m}{\sigma^2 \surd(2c_n)} (\hat{\sigma}^2 - \sigma^2) + b(\hat{\sigma}^2) \frac{\hat{\sigma}^2}{\sigma^2} \\ &= \frac{p-m}{\sigma^2 \surd(2c_n)} (\hat{\sigma}^2 - \sigma^2) + b(\sigma^2) + \frac{1}{\sigma^2} \{ b(\hat{\sigma}^2)\hat{\sigma}^2 - b(\sigma^2)\sigma^2 \}. \end{aligned}$$

Since $\hat{\sigma}^2 \xrightarrow{\mathcal{P}} \sigma^2$, by continuity, the second expression tends to 0 in probability and the conclusion follows from theorem 1.

A.3. Proof of theorem 3

By lemma 2.2 of Wang and Yao (2013), we have

$$G_n(x) = \sum_{i=1}^{p} \lambda_{n,i} - p \int x \, \mathrm{d}F_{c_n, H_n}(x) \xrightarrow{\mathcal{D}} \mathcal{N}\{m(x), v(x)\}.$$

The asymptotic mean does not change for non-Gaussian data, $m(x) = 0$, but the asymptotic variance is $v(x) = c\gamma\sigma^4$.

A.4. Proof of theorem 5

We prove that $\lim_{N,T\to\infty} P\{\mathrm{PC}^*(m) < \mathrm{PC}^*(\bar{m})\} = 0$ for all $m \neq \bar{m}$ and $m \leqslant m_0$. By definition,

$$\begin{aligned}
&\mathrm{PC}^*(m) - \mathrm{PC}^*(\bar{m}) < 0 \\
&\quad\Leftrightarrow \hat{\sigma}^2_*(m) - \hat{\sigma}^2_*(\bar{m}) < (\bar{m} - m)\, \hat{\sigma}^2_*(m_0)\, g(N,T) \\
&\quad\Leftrightarrow \hat{\sigma}^2_*(\bar{m}) - \hat{\sigma}^2_*(m) > (m - \bar{m}) \hat{\sigma}^2_*(m_0)\, g(N,T).
\end{aligned}$$

Consider first $m < \bar{m}$. We have by expression (7)

$$\hat{\sigma}^2_*(m) - \hat{\sigma}^2_*(\bar{m}) = \{\hat{\sigma}^2(m) - \hat{\sigma}^2(\bar{m})\}\{1 + o_p(1)\}.$$

Moreover,

$$\begin{aligned}
(N-m)\{\hat{\sigma}^2(m) - \hat{\sigma}^2(\bar{m})\} &= \sum_{m < i \leqslant \bar{m}} \lambda_i - (\bar{m} - m)\, \hat{\sigma}^2(\bar{m}) \\
&\geqslant (\bar{m} - m)\{\lambda_{\bar{m}} - \hat{\sigma}^2(\bar{m})\}.
\end{aligned}$$

Since $\lambda_{\bar{m}} \to \sigma^2\{\alpha_{\bar{m}}/\sigma^2 + 1 + c(1 + \sigma^2/\alpha_{\bar{m}})\}$ and $\hat{\sigma}^2(\bar{m}) \to \sigma^2$ (in probability), the lower bound above converges to $(\bar{m} - m)\sigma^2\{\alpha_{\bar{m}}/\sigma^2 + c(1 + \sigma^2/\alpha_{\bar{m}})\}$ which is positive. The conclusion $P\{\mathrm{PC}^*(m) < \mathrm{PC}^*(\bar{m})\} \to 0$ will follow if the penalty satisfies

$$(N-m)\, g(N,T) < \frac{\sigma^2}{\hat{\sigma}^2_*(m_0)} \left\{ \frac{\alpha_{\bar{m}}}{\sigma^2} + c\left(1 + \frac{\sigma^2}{\alpha_{\bar{m}}}\right) \right\}, \tag{20}$$

for large N and T. By assumption $\alpha_{\bar{m}}/\sigma^2 > \sqrt{c}$ which implies that $\alpha_{\bar{m}}/\sigma^2 + 1 + c(1 + \sigma^2/\alpha_{\bar{m}}) > c + 2\sqrt{c}$. However, we have that $\sigma^2/\hat{\sigma}^2_*(m_0) = 1 + \beta/T + o_p(1/T)$ where β is some constant (depending on m_0, c and σ^2). So with the $g(N,T)$ in equation (13), we have $(N-m)\, g(N,T) = (c + 2\sqrt{c})(1 + T/N^{1+\delta}) \ll (c + 2\sqrt{c})\{1 + \beta/T + o_p(1/T)\}$ and the conclusion follows. Next, consider the case where $m > \bar{m}$. We have

$$\begin{aligned}
(N-m)\{\hat{\sigma}^2(\bar{m}) - \hat{\sigma}^2(m)\} &= \sum_{\bar{m} < i \leqslant m} \lambda_i - (m - \bar{m})\, \hat{\sigma}^2(\bar{m}) \\
&\to (m - \bar{m})\, \sigma^2(c + 2\sqrt{c}), \qquad \text{in probability,}
\end{aligned}$$

because $\lambda_i \to \sigma^2(1 + \sqrt{c})^2$ in probability for $\bar{m} < i \leqslant m$. Note that $\hat{\sigma}^2_*(m_0) \to \sigma^2$ in probability and, with the $g(N,T)$ in equation (13), we have

$$\liminf_{N,T\to\infty} (N-m)\, g(N,T) \geqslant c + 2\sqrt{c}.$$

The conclusion follows.

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Supporting information

Additional 'supporting information' may be found in the on-line version of this article:

'On estimation of the noise variance in high-dimensional probabilistic principal component analysis (supplementary report)'.